ZIGURAT
INSTITUTE OF TECHNOLOGY

MODULE 7
Unit 3

AI Applied to BIM Project Management and Cost Estimation

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My Friend Paul.

"£65 Per Square Foot."

He asked a few questions.
He gave the rate.
He is an Estimator.

Five months later, the build came in at £65.23 per square foot.

£

65

per square foot

Why didn't Paul provide a total price?

"£65 Per Square Foot"

A) He didn't know.
B) The rate is what he knew, the total wasn't.
C) He didn't want to be wrong.
D) Other

Objectives: What we are covering

STEP 01
Methods of Modeling (for quantities)

STEP 02
Build Modeling Spec with AI

STEP 03
Model Autochecking

STEP 04
Create Estimating dashboard with API

What this class is NOT:

- We are not turning you into estimators.
- We are not teaching you to build the model.
- We are not trusting every model quantity automatically.

Estimating is expert judgment

Good estimators do not just count things.

They recognise:

Patterns	**Risks**	**Methods**
Labour	**Materials**	**Assumptions**

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The Problem: AI Estimation Illusion

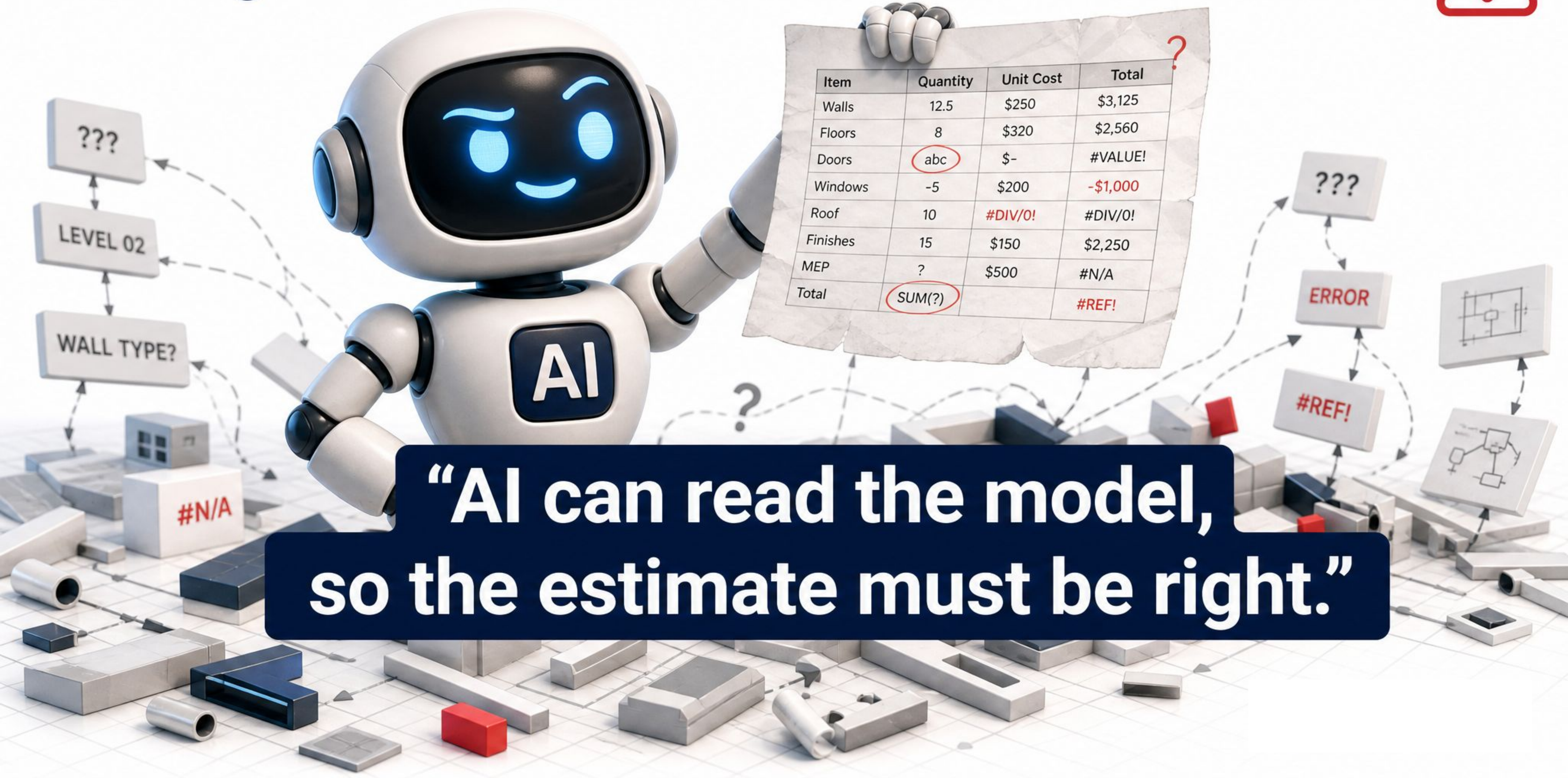
Why AI Estimating Fails Without Model Quality

AI's ability to read, summarise, and report model-based data is powerful, but it creates a **dangerous illusion**: a professional-looking estimate does not guarantee reliability.

> ***"AI can make a bad model sound confident. That does not make the model useful."***

- AI generates estimates from model-based data efficiently.
- Confidence in presentation ≠ Accuracy in data.

The Big Mistake.



“AI can read the model, so the estimate must be right.”

On real projects, there is never one model.

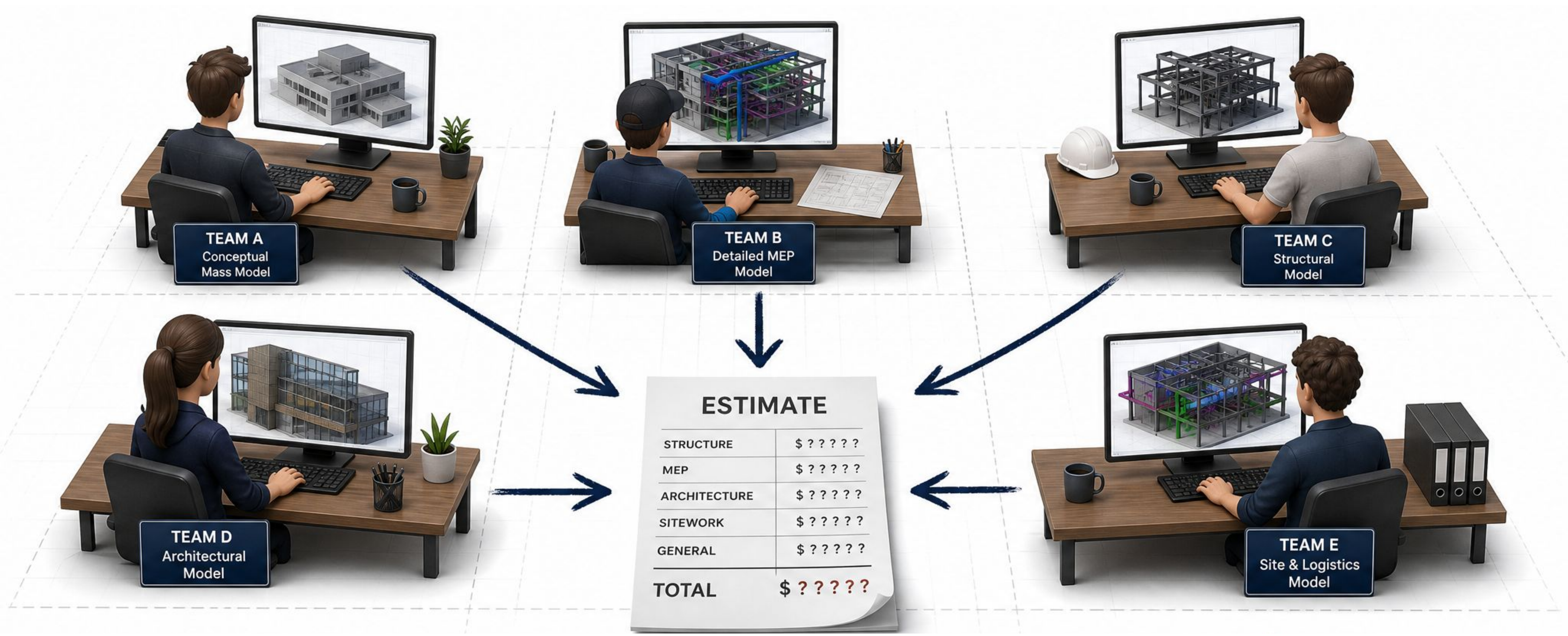
Team A
MEP
own model • own conventions

Team B
Structural
own model • own conventions

Team C
Facade
own model • own conventions

Team D
Architecture
own model • own conventions

All feeding one estimate.



Multiple project teams creating different models, all feeding one estimate.

Multiple teams, one estimate.

Different models, different conventions. One number expected at the end.

- **Architecture**
- **Structural**
- **MEP**
- **Facade**

ESTIMATE

Interior walls	£420,000
Structural frame	£1.2M
MEP	£860,000
Facade	£1.05M

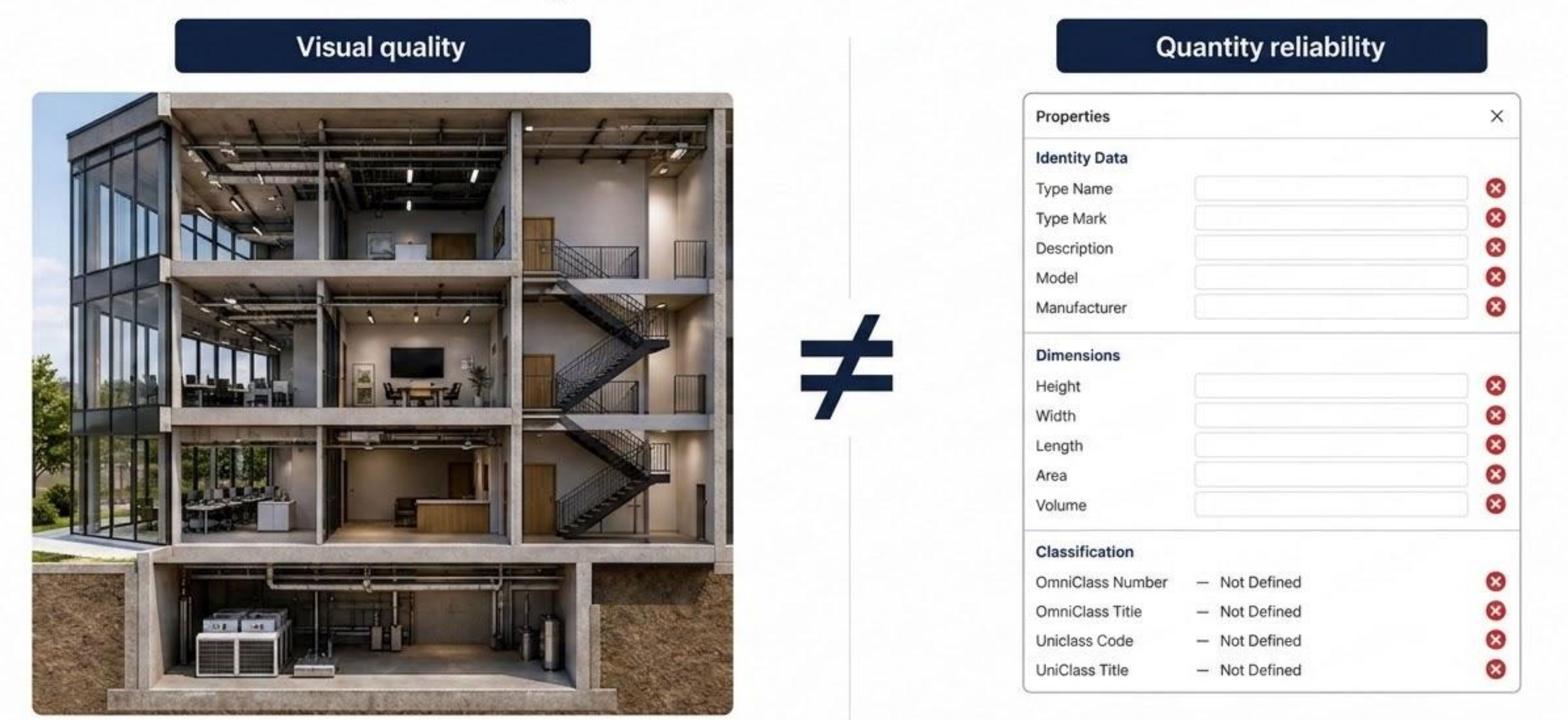
What we are building:

A trusted workflow between the model and the estimate.

Requirements › **Contract** › **Verification** › **Dashboard** › **Approval** › **Estimate**

Six stages. Repeat this phrase three times across the talk.

A good-looking model is not always a measurable model.



What makes model quantities trustworthy?

01. Foundation
Purpose
Defining why the model is being created.

02. Scope
Requirements
Specifying what data is needed.

03. Logic
Rules
Establishing standards for data entry.

04. Quality
Verification
Automated and manual checks.

05. Validation
Approval
Sign-off from key stakeholders.

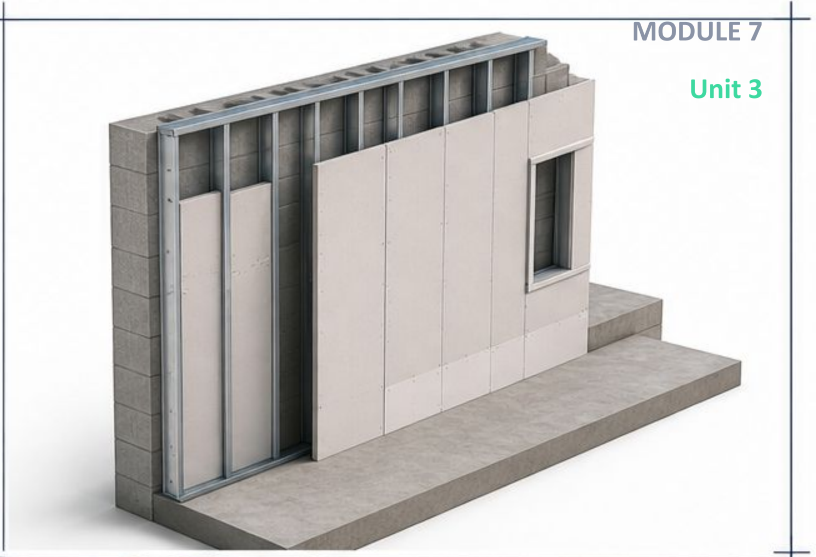
06. Execution
Use
Applying trustworthy data to estimates.

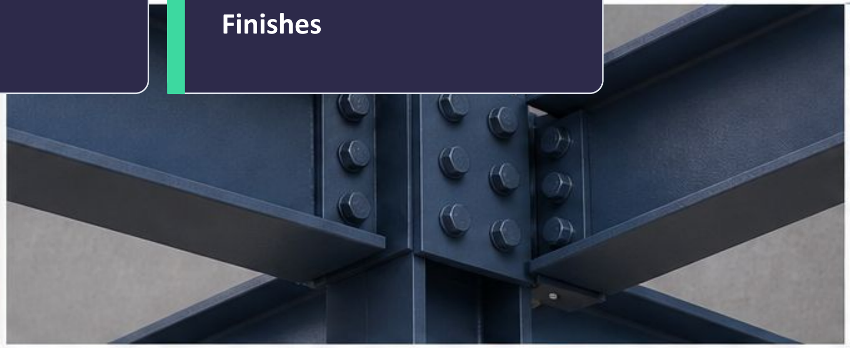
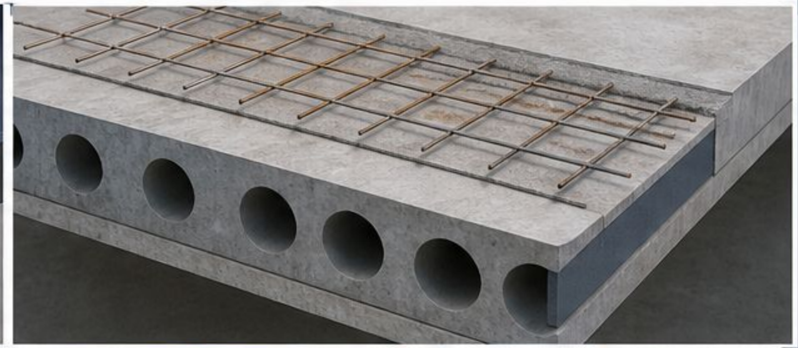
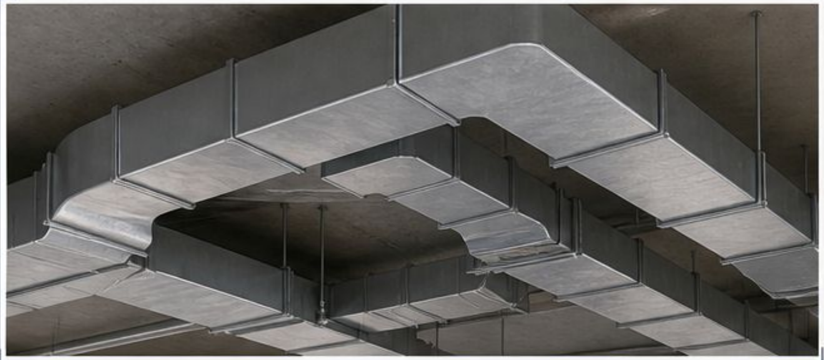
A structured workflow ensures that model quantities are reliable for project estimation.

Methods of measurement matter.

Different elements need different measurement logic.

- **Reinforcement**
- **Ductwork**
- **Steel sections**
- **Precast panels**
- **Wall assemblies**
- **Finishes**



Methods of measurement matter.

Different elements need different measurement logic.

Geometric Logic

Calculations based on physical dimensions, volumes, and surface areas of structural components.

System Logic

Measurement based on system performance, linear runs of MEP, or functional capacities.

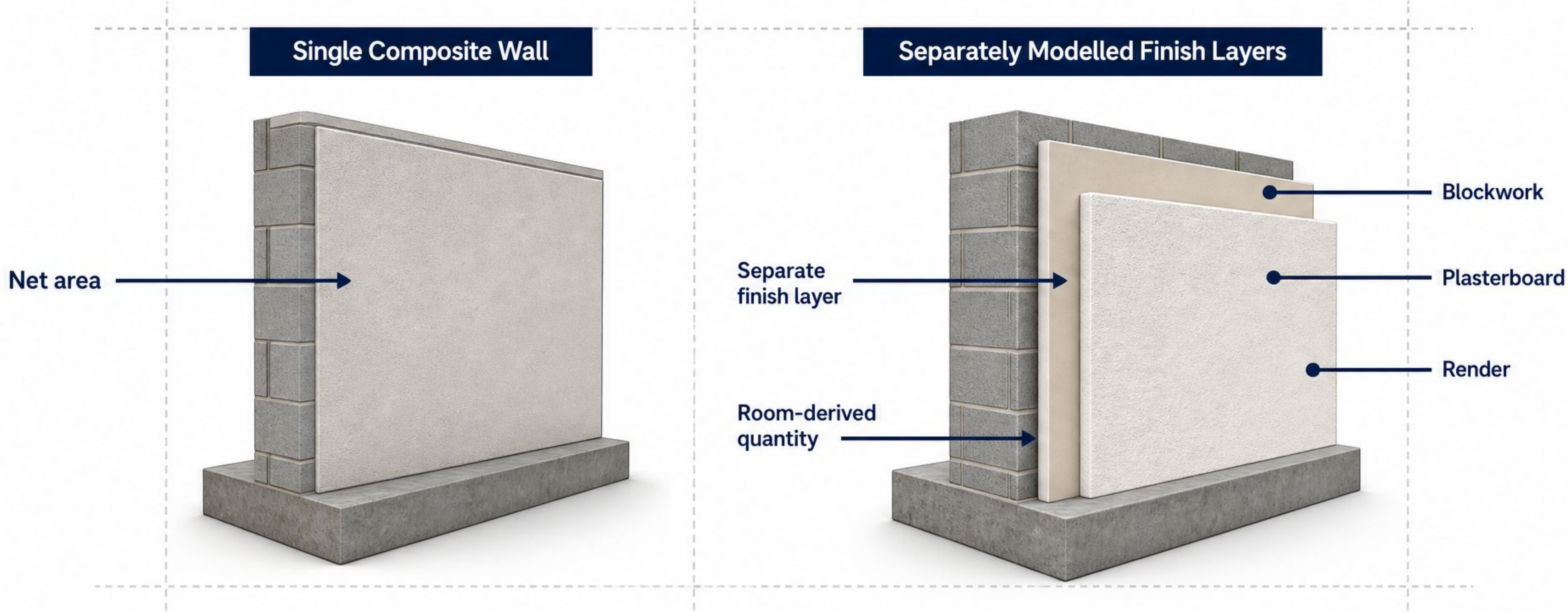
Property Logic

Quantities derived from metadata and parameters attached to non-geometric model elements.

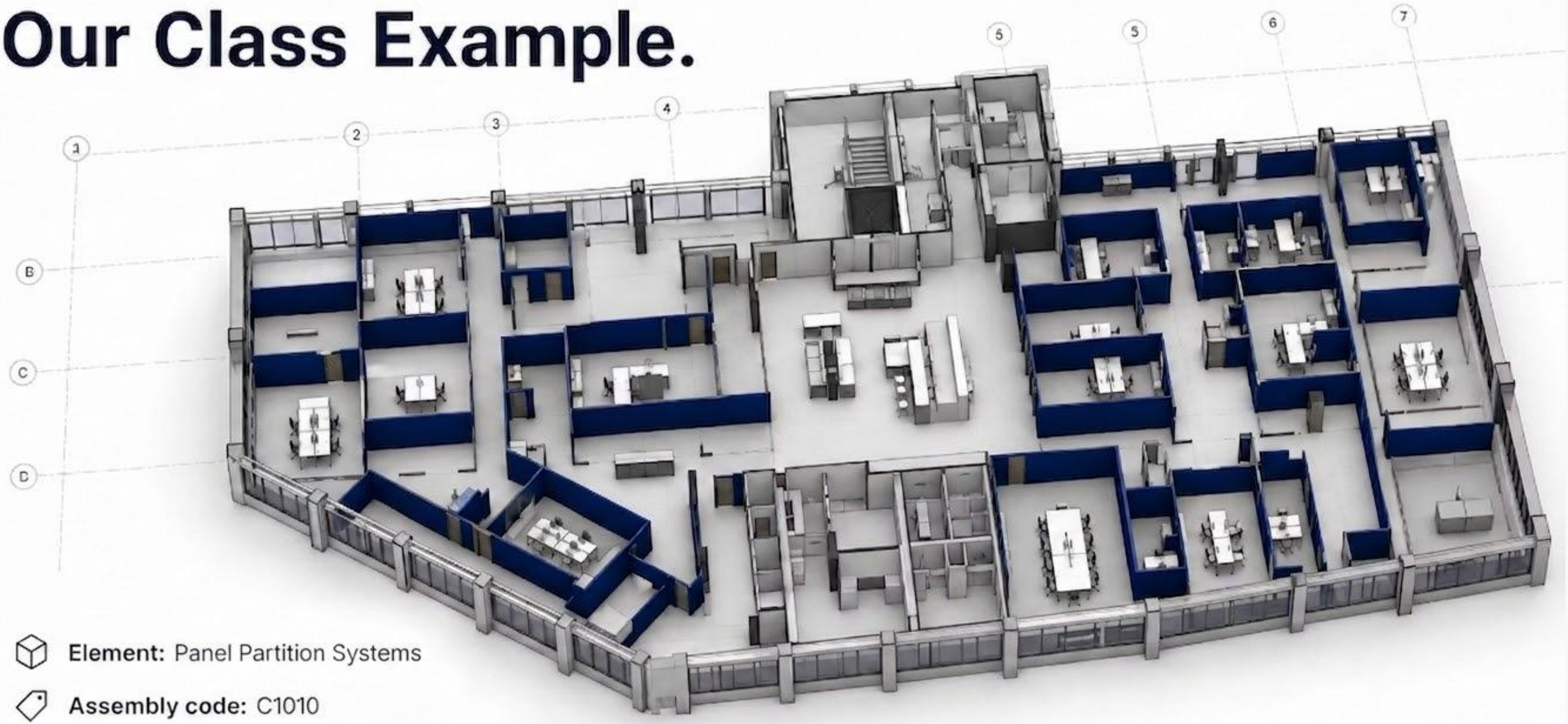
Applying the correct measurement ruleset is vital for accurate project estimates.

Example: Wall Layers and Finishes

Measurement depends on how the element is modelled.

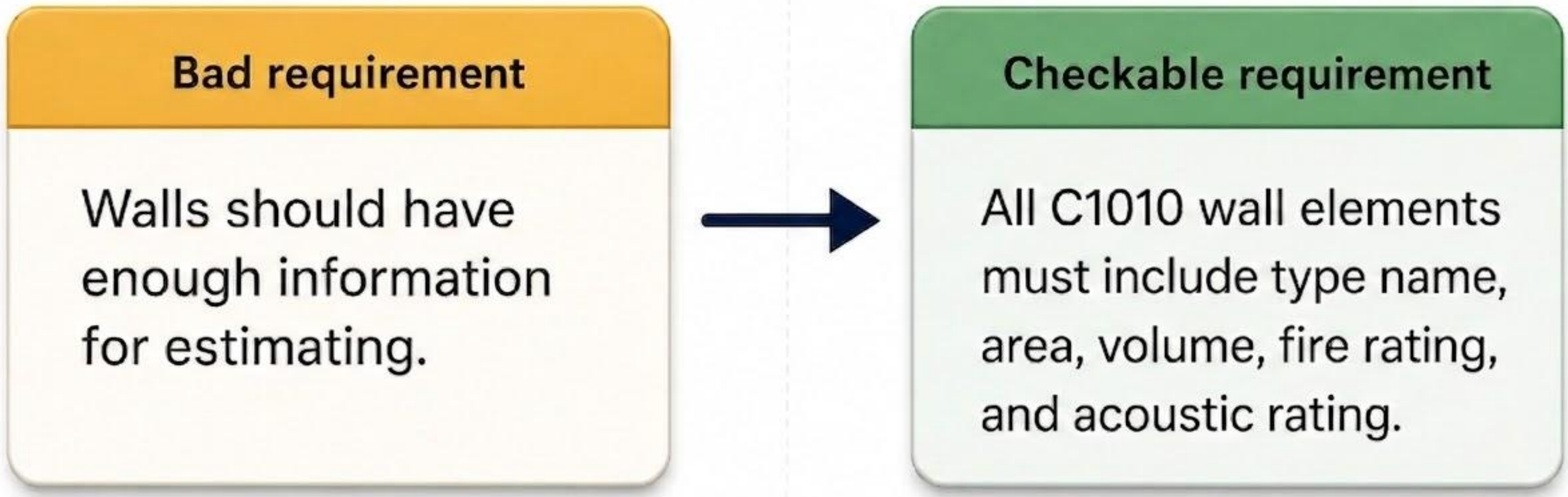


Our Class Example.



- **Element:** Panel Partition Systems
- **Assembly code:** C1010
- **Scope:** Interior walls
- **Verified elements:** 81

From Requirement to Check.



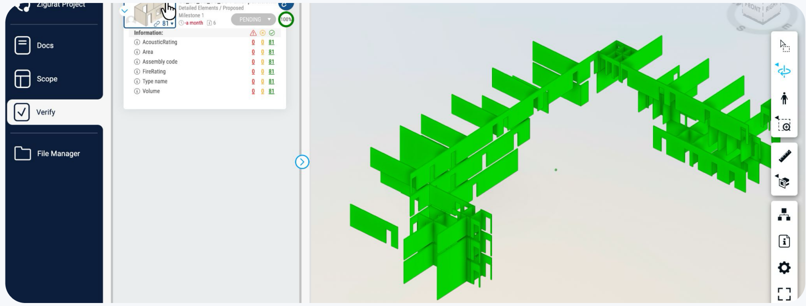
Exercise 1: Define 'Good Enough'.

For C1010 walls to support estimating, what must be true?

- [ ] Good1
- [ ] Good2
- [ ] Good3
- [ ] Good4
- [ ] Good5

5 minutes. Every line must be checkable.

Required Model Data.



For C1010 walls, we need:

- **Assembly Code** ✓
- **Type Name** ✓
- **Area** ✓
- **Volume** ✓
- **Fire Rating** ✓
- **Acoustic Rating** ✓

Not every check is the same kind.

Automated

Rules check if specific properties exist within the BIM model automatically.

Manual

Professional review to ensure the modelling method is appropriate for the use case.

AI-Assisted

Leverage AI to summarise issues and draft comprehensive review notes.

Where else AI helps in the workflow.

Not just checking the model. AI also helps us build the spec.

AI

Generate a description

AI drafts the project or element description from your brief, so Purpose and Scope start with something real.

AI

Generate checklists

For the checks that cannot be automated, AI writes a structured manual checklist your reviewer can run through.

AI

Document set on each task

AI links the right document set to each requirement and task in Scope so the contract is traceable.

AI

Define the assembly code rule

AI generates the verification rule for assembly code from a plain-English description of what you want to enforce.

AI

Draft a doc section for your element

In Documents, AI writes a section specific to the chosen element so the spec document is element-aware.

AI

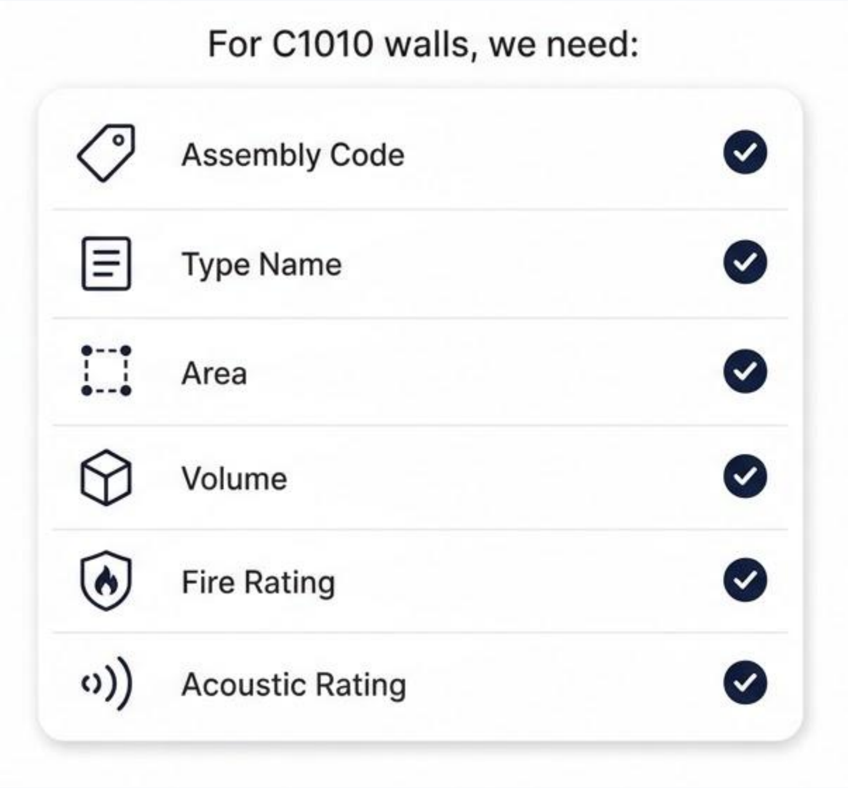
Check your technical writing

AI reviews your spec for ambiguity, inconsistency, and unenforceable language. The kind of red-pen pass that takes hours by hand.

AI organises and drafts. You decide what stays.

81 Verified Wall Elements.

These are the elements allowed into the estimating workflow.



81 ✓

verified wall elements

Model quality is now a result.

Not an opinion. Not a screenshot. Not a bad promise.

Requirements | Contract | Verification | Dashboard | Approval | Estimate

End of Session 1

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NEXT

Using AI + API for Model-Based Estimating…